

LAB EXPERIMENTS

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24) Python Program – Alphabet Inc. Financial Data Line Chart

Code :

```
import pandas as pd
import matplotlib.pyplot as plt

# Read the CSV file
data = pd.read_csv("fdata.csv")

# Convert Date column to datetime
data["Date"] = pd.to_datetime(data["Date"])

# Select data from October 3, 2016 to October 7, 2016
data = data[
    (data["Date"] >= "2016-10-03") &
    (data["Date"] <= "2016-10-07")
]

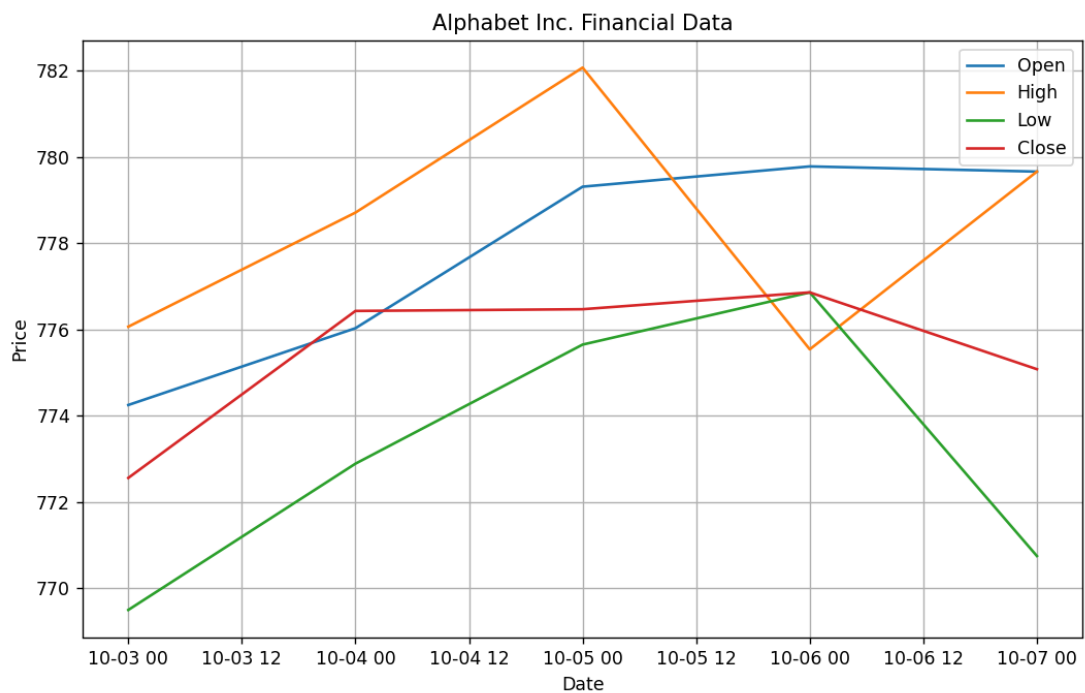
# Plot financial data
plt.figure(figsize=(10, 6))
```

```
plt.plot(data["Date"], data["Open"], label="Open")
plt.plot(data["Date"], data["High"], label="High")
plt.plot(data["Date"], data["Low"], label="Low")
plt.plot(data["Date"], data["Close"], label="Close")
```

```
plt.xlabel("Date")
plt.ylabel("Price")
plt.title("Alphabet Inc. Financial Data")
plt.legend()
plt.grid(True)
```

```
plt.show()
```

OUTPUT :



```
● PS E:\Query process> e;; cd 'e:\Query process'; & 'C:\Users\V  
bundled\libs\debugpy\launcher' '64575' '--' 'E:\Query process\  
E:\Query process\import pandas as pd.py:8: UserWarning: Could  
please specify a format.  
data["Date"] = pd.to_datetime(data["Date"])
```

25) Python program to plot two or more lines with legends, different widths and colors.

Code :

```
import matplotlib.pyplot as plt

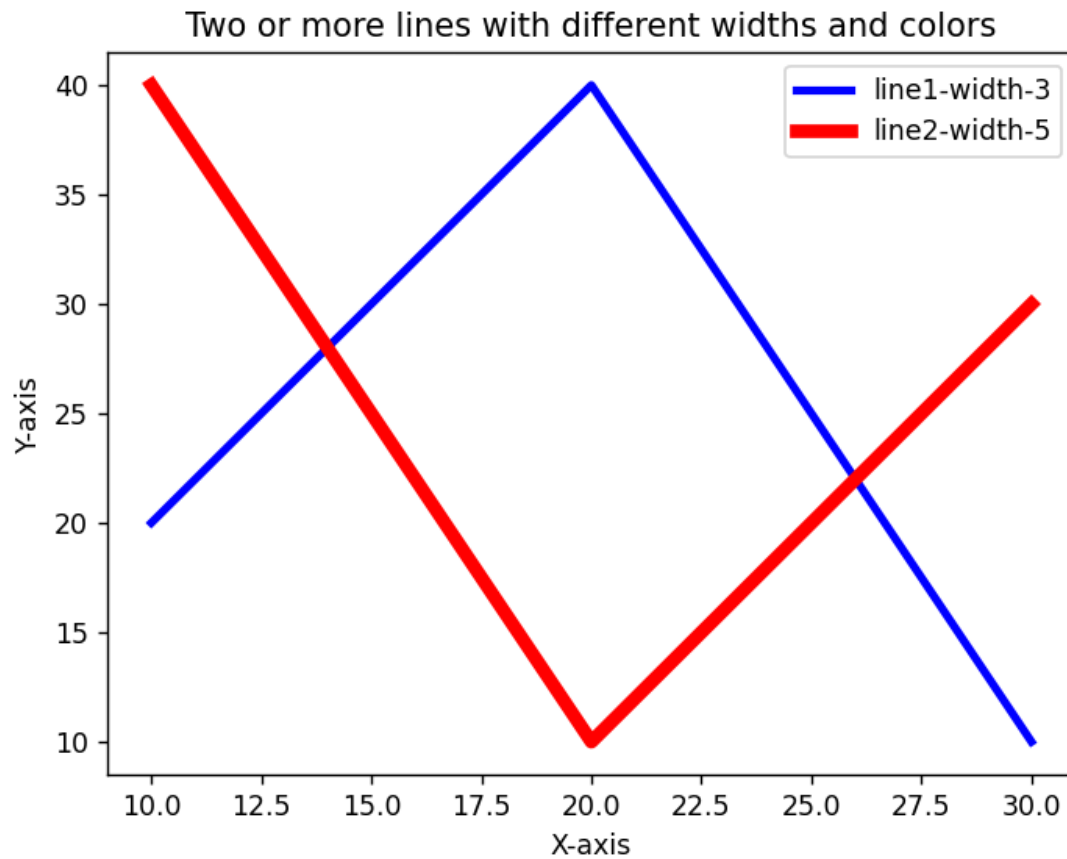
# X and Y values
x = [10, 20, 30]

y1 = [20, 40, 10]
y2 = [40, 10, 30]

# Plot two lines with different colors and widths
plt.plot(x, y1, color="blue", linewidth=3, label="line1-width-3")
plt.plot(x, y2, color="red", linewidth=5, label="line2-width-5")
# Add title and labels
plt.title("Two or more lines with different widths and colors")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
# Display legend
plt.legend()

# Display graph
plt.show()
```

OUTPUT :



26) Write a Python program to create multiple plots.

Code :

```
import matplotlib.pyplot as plt
```

```
# Data
```

```
x = [1, 2, 3, 4, 5]
```

```
y1 = [10, 20, 15, 25, 30]
```

```
y2 = [5, 15, 20, 18, 25]
```

```
y3 = [30, 25, 20, 15, 10]
```

```
y4 = [10, 15, 25, 20, 30]
```

```
# Create one figure with 4 plots
```

```
plt.figure(figsize=(10, 7))
```

```
# Graph 1
```

```
plt.subplot(2, 2, 1)
```

```
plt.plot(x, y1)
```

```
plt.title("Line Plot")
```

```
plt.xlabel("X-axis")
```

```
plt.ylabel("Y-axis")
```

```
# Graph 2
```

```
plt.subplot(2, 2, 2)
```

```
plt.bar(x, y2)
```

```
plt.title("Bar Plot")
```

```
plt.xlabel("X-axis")
```

```
plt.ylabel("Y-axis")
```

```
# Graph 3
```

```
plt.subplot(2, 2, 3)
```

```
plt.scatter(x, y3)
```

```
plt.title("Scatter Plot")
```

```
plt.xlabel("X-axis")
```

```
plt.ylabel("Y-axis")
```

```
# Graph 4
```

```
plt.subplot(2, 2, 4)
```

```
plt.plot(x, y4, marker="o")
```

```
plt.title("Line Plot 2")
```

```
plt.xlabel("X-axis")
```

```
plt.ylabel("Y-axis")
```

```
# Adjust spacing
```

```
plt.tight_layout()
```

```
# Display all 4 graphs in one figure
```

```
plt.show()
```

OUTPUT :

